

Laser Safety

Policy Number	POL-SAF-072	Owner / Department	Perioperative Services / Safety
Effective Date	2023-10-15	Version	2.1
Last Revised	2026-02-20	Next Review	2028-02-20
Approved By	Environment of Care Committee	Applies To	All personnel who operate, assist with, or are present during the use of surgical and therapeutic lasers in the OR, procedure rooms, and clinics

1. Purpose and Scope

This policy governs the safe use of Class 3B and Class 4 health-care lasers to prevent eye and skin injury, airway and surgical fires, and plume-inhalation hazards. It follows **ANSI Z136.3** (Safe Use of Lasers in Health Care) and AORN perioperative guidance.

A designated **Laser Safety Officer (LSO)** administers the program, approves operators, and defines the **Nominal Hazard Zone (NHZ)** for each device.

2. Controls and Protective Measures

- Wavelength-specific protective eyewear with the correct optical density is worn by everyone in the NHZ; the patient's eyes are protected per wavelength.
- Controlled-access signage and a lit warning sign are posted at every entrance to the NHZ; windows are covered when required.
- The laser is kept in **STANDBY** except during active firing; only the operator activates emission, on the surgeon's command.
- A smoke-evacuation system captures surgical plume at the source; high-filtration respiratory protection is used where indicated.
- Fire-risk controls apply during airway and head/neck cases: minimize supplemental oxygen, use laser-resistant endotracheal tubes, and keep saline and wet sponges available.

IMPORTANT — If a surgical or airway fire ignites, **STOP gas flow and the laser, remove burning material, and extinguish immediately (for an airway fire, remove the tube and disconnect oxygen).** Then treat the patient and activate the fire response.

3. Operation and Credentialing

1. Verify operator credentialing and device-specific training before the case.
2. Complete a pre-use safety time-out: confirm eyewear, signage, NHZ boundaries, plume evacuation, and fire precautions.
3. Test-fire the laser into an appropriate target, confirm settings, and keep it in STANDBY between activations.

4. Account for all protective measures at case end and document laser use in the operative record.

4. Severity & Reporting

Classify and notify per **POL-RM-013**. A surgical or airway fire causing serious injury, or a laser eye injury, is a **SEV-1**; a minor burn or an unintended tissue injury without serious harm is a **SEV-2**; a control lapse without injury is a **SEV-3 / Near Miss**.

Severity	Notify	Deadline
SEV-1 (catastrophic / sentinel)	LSO, OR charge, Risk Management, administrator on call	Within 1 hour
SEV-2 (major)	LSO and perioperative manager	Within 4 hours
SEV-3 / Near Miss (minor)	LSO via electronic event report	Within 24 hours

If the event involved a laser-device malfunction, also follow **POL-SAF-073**; staff plume or burn exposures are cross-referenced to **POL-EH-001**.

5. References

- ANSI Z136.3 — Safe Use of Lasers in Health Care.
- AORN Guideline for Laser Safety.
- POL-RM-013 — Incident Reporting and Event Severity Classification.
- POL-SAF-073 — Medical Device Malfunction, Recall, and FDA Reporting.
- POL-EH-001 — Bloodborne Pathogen Exposure and Needlestick.